

GENERATE LESS, CLASSIFY MORE: PROBE-GUIDED PRUNING AND DISCRIMINATIVE HEADS FOR EFFICIENT INTENT DETECTION AND SLOT FILLING WITH SMALL LANGUAGE MODELS

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ABSTRACT

Task-oriented dialogue increasingly relies on small language models (SLMs) that emit structured output (intent + slots) via autoregressive generation. We show that for *closed-set* intent detection and slot filling this is both slow and brittle, and unnecessary. We present a recipe that converts a generative SLM into a **discriminative single-pass** model: (1) a frozen linear-probe sweep over a small predefined depth grid $\{L/4, L/2, 3L/4, L\}$ identifies an aggressive candidate pruning region *a priori*; (2) a linear intent head and a token-level slot head (softmax or CRF) replace token-by-token generation. On a 124M-parameter decoder, pruning to 3 of 12 layers yields approximately 5 ms P50 CPU latency versus approximately 359 ms for the autoregressive baseline on ATIS ($\sim 70\times$ speedup). This factorizes into $\sim 18\times$ from replacing autoregressive decoding with a single discriminative pass and $\sim 3.8\times$ from depth pruning, so most of the win comes from the architectural conversion rather than from pruning. The discriminative heads also remove a generative parse-failure rate of up to 46.7%: a closed label space cannot emit an unparseable output, so validity is structural rather than empirical. At a matched 3,000-example budget, depth 3 matches depth 12 at 96.08% intent accuracy on ATIS while reducing P50 latency from 20.4 to 5.4 ms. Fine-tuning at candidate depth 3 preserves most downstream task performance across ATIS, SNIPS, MASSIVE, CLINC150, and BANKING77 while substantially reducing inference latency relative to full depth. A diagnostic dense sweep (layers 1–12) reveals that linear accessibility of intent information precedes fine-tuning sufficiency by several layers—a cheap intent probe alone under-selects depth on the shallow end, and a token-level slot probe on SNIPS mirrors that under-selection. We further report a negative-results study of CPU inference optimizations: dynamic INT8 and `torch.compile` give no benefit for shallow models, while ONNX Runtime yields a $4.5\times$ speedup at the cost of brittle export.

1 INTRODUCTION

Small language models are increasingly deployed for task-oriented natural language understanding (NLU), where a user utterance is mapped to a discrete *intent* and a set of typed *slots*. A common pattern is to fine-tune a generative SLM to emit a structured object (e.g. JSON `{"intent": ..., "slots": ...}`) autoregressively. This inherits two costs from generation: (i) latency that scales with output length, and (ii) a nonzero rate of *malformed* outputs that fail to parse into the target schema.

We argue that closed-set intent + slot filling does not need generation. Our central claim: *converting a generative SLM into a probe-pruned discriminative single-pass model yields a large CPU speedup and zero structured-output parse failures. Frozen probes identify an aggressive candidate pruning region a priori, but linear accessibility of task information precedes fine-tuning sufficiency by several layers; the final deployed depth is confirmed by downstream fine-tuning, not by the probe alone.*

Contributions. Two primary contributions: **(C1) Probe-guided candidate pruning:** a frozen linear probe over a small predefined depth grid identifies an aggressive candidate region before fine-tuning, reducing the need to exhaustively train every depth. Dense diagnostics further reveal a failure boundary where linear task accessibility precedes fine-tuning sufficiency; on the slot-bearing SNIPS benchmark a token-level BIO-tag probe exposes this gap. **(C2) Generative $\rightarrow$ discriminative conversion:** a large speedup, decomposed into its decoding and pruning components, plus the elimination of a measured 46.7% generative parse-failure rate. We note that zero parse failures follows from the closed output parameterization rather than being an empirical result; the substantive number is the generative failure rate it displaces. Supporting analyses cover a measured implicit-slot scope statement (0% on our benchmarks) that bounds where span-based heads suffice, and a study of common CPU inference optimizations that shows the systems win comes from architecture (prune + classify), not post-hoc kernel/quant tricks. We position our contribution honestly: against the generative baseline the win is decisive; against a purpose-built bidirectional encoder we are competitive, and our recipe requires no separate encoder—it reuses an existing decoder.

Why a decoder? Our target setting is one in which an existing causal decoder is already available and must be adapted for latency-sensitive closed-set NLU. The goal is therefore not to outperform purpose-built encoders, but to determine how much of an existing decoder is actually necessary and to eliminate autoregressive decoding when the output space is closed and structured.

2 RELATED WORK

Structured NLU. Discriminative intent + slot prediction has a long lineage; JointBERT (Chen et al., 2019) and successors fine-tune an encoder (Devlin et al., 2019; Vaswani et al., 2017) with an intent classifier and a token-level tagger. Constrained decoding (Willard & Louf, 2023) restores validity for generative structured outputs but retains autoregressive cost; discriminative heads make validity structural at no decode cost.

Efficient transformer inference. Layer dropping (Fan et al., 2020), dynamic early exit (Xin et al., 2020), distillation (Sanh et al., 2019), and rotation-based quantization (Ashkboos et al., 2024) all target inference cost. Closer to our setting, recent decoder-only depth-pruning work (Men et al., 2024; Gromov et al., 2024) removes LLM layers using an *unsupervised* block-similarity or influence signal, then heals the result with light fine-tuning; we instead use a *supervised* linear probe over task labels on a small predefined depth grid, and—unlike that line of work—explicitly test and find cases where the probe signal itself is misleading (Section 5). We use *static* depth pruning guided by frozen probes, complementary to and stackable with kernel/quant work (§6). The upstream trend toward small language models (Allal et al., 2024; Abdin et al., 2024) makes the “how much of the decoder do we actually need” question increasingly practical for on-device NLU.

Probing. Prior probing (Alain & Bengio, 2017; Tenney et al., 2019) primarily analyzes *where* information is encoded in a pretrained network. We instead operationalize frozen probing as a pre-fine-tuning signal for static decoder truncation, and characterize an important limitation: linear task-information accessibility can precede downstream fine-tuning sufficiency at extreme shallow depths, so a probe alone does not identify the deployed depth. A substantial literature already cautions against reading probe accuracy at face value: it can reflect probe capacity rather than representation content (Hewitt & Liang, 2019), ignore extraction difficulty (Voita & Titov, 2020), or decode information the model does not causally use (Elazar et al., 2021); see Belinkov (2022) for a survey. Our failure mode is distinct from all of these and complementary to them: we hold the probe fixed and ask whether its accuracy at layer ℓ predicts the accuracy of a model *truncated at ℓ and then fine-tuned*—a deployment-time question that decodability, extraction cost, and causal-use critiques do not address. We find it does not, with negative within-dataset rank correlation on four of five benchmarks. Concurrent work (Ibañez-Lissen et al., 2025) similarly uses frozen probes before fine-tuning to pick a layer-pruning cutoff, but in a different domain (source-code vulnerability classification), probing a proxy structural signal (code-complexity metrics) rather than the task label itself, via a dense per-layer sweep rather than a predefined sparse grid with a tolerance rule; critically, that work treats probe accuracy as validated evidence for the cutoff, whereas we show probe accessibility can *over-recommend* an aggressively shallow depth that then underperforms after fine-tuning.

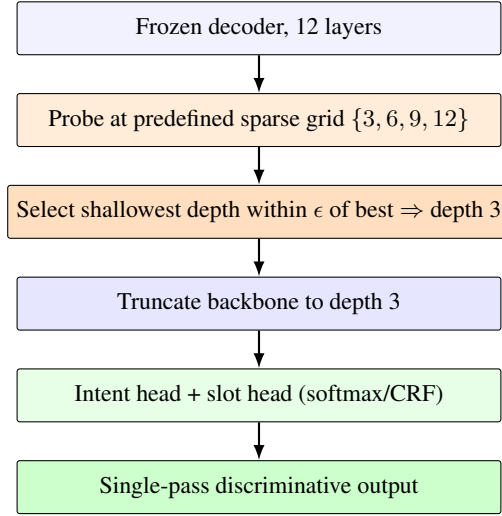


Figure 1: Pipeline (simplified causal flow). A frozen decoder is probed at a *predefined sparse* candidate grid $\{L/4, L/2, 3L/4, L\}$ ($= \{3, 6, 9, 12\}$ for $L=12$); the shallowest depth within ϵ of the best probe accuracy is selected *a priori* as the candidate pruning depth. The backbone is then truncated to that depth and fine-tuned with a linear intent head over the pooled final state and a token-level slot head (softmax or CRF). No autoregressive decoding at inference. A separate dense sweep (layers $1-L$) is used diagnostically in Section 5 to test the rule’s limits and expose the gap between linear accessibility and fine-tuning sufficiency.

3 METHOD

Setup. A closed intent set $\mathcal{I}$ and BIO slot tags $\mathcal{T}$; we predict one intent and a tag per token in a single forward pass. **(3.1) Probe-guided candidate pruning.** We freeze the pretrained decoder and, in one forward pass with hidden-state output, mean-pool the hidden states at a *predefined sparse candidate grid* of depths $\{L/4, L/2, 3L/4, L\}$ over non-pad tokens. A linear probe (logistic regression) is trained on each depth’s pooled features to predict the intent. We select the shallowest depth whose probe accuracy is within ϵ of the best (we use $\epsilon = 0.02$). *This grid is chosen a priori, not tuned to the data.* A dense per-layer sweep ($\{1, \dots, L\}$) is used as a diagnostic in Section 5 to test the rule’s limits, not as the deployment procedure. The probe is therefore used to identify a candidate pruning region rather than to guarantee the final downstream optimal depth. **(3.2) Discriminative heads.** We truncate the backbone to the selected depth, add a linear intent head over mean-pooled hidden states and a per-token slot head (softmax cross-entropy, or a linear-chain CRF (Lafferty et al., 2001) with Viterbi decoding). The joint loss is $\mathcal{L} = \mathcal{L}_{\text{intent}} + \lambda \mathcal{L}_{\text{slot}}$ with $\lambda = 2$; we length-normalize the CRF NLL so it does not dominate the intent term. **(3.3) Reliability.** Because output is a fixed set of classifier decisions, it is structurally valid: parse-failure rate is 0 by construction.

4 EXPERIMENTAL SETUP

Datasets. Table 1 lists the five public benchmarks. **Models.** Backbone: gpt2-124M (Radford et al., 2019); encoder baseline: DistilBERT (Sanh et al., 2019); all runs use HuggingFace Transformers (Wolf et al., 2020) on top of PyTorch (Paszke et al., 2019). **Metrics.** Intent accuracy, span-level slot F1, parse-failure rate, and single-example P50 latency on a commodity CPU (Apple Silicon arm64, 10 threads). For fairness across the architecture comparison we cap training-set size at 5,000 examples for both the encoder baseline and the pruned model (a hard slice of the shuffled split, applied identically to both architectures); datasets whose full training split is smaller than this cap use their entire split rather than being upsampled. ATIS’s split (4,274 examples, Table 1) falls below the cap, so both architectures train on all 4,274 ATIS examples. The generative baseline is bounded to 3000 train / 300 eval for CPU-tractable autoregressive decoding.

Table 1: Datasets. Intent-only benchmarks stress fine-grained classification (150 and 77 classes); slot-filling benchmarks stress structured output. All are public; train and test sizes are shown. Counts are measured directly from the exact files our pipeline loads (matching `n_train/n_test` in every released result JSON), not copied from prior papers; our ATIS and SNIPS mirrors differ slightly from other commonly-cited reformatting of these benchmarks (e.g. ATIS is often cited at 4,478 train / 893 test elsewhere in the literature). ATIS #Slots (135) is the current BIO tag-label count (both B-/I- registered per observed slot type); the M1 encoder/generative baseline runs reported in §5 were produced immediately before this tag-vocabulary convention was adopted and used 101 labels for that run only, which does not affect their reported accuracy/F1 figures.

Dataset	#Intents	#Slots	Train	Test	Used for
ATIS (Hemphill et al., 1990)	26	135	4,274	586	C1, C2, C3, C4
SNIPS (Coucke et al., 2018)	7	72	13,784	700	C1, C2, C3
MASSIVE-en (FitzGerald et al., 2023)	60	55	11,514	2,974	C1, C2, C3
CLINC150 (Larson et al., 2019)	150	–	15,000	4,500	C1, C2
BANKING77 (Casanueva et al., 2020)	77	–	10,003	3,080	C1, C2

Table 2: Main comparison on ATIS. The pruned discriminative model is $\sim 70\times$ faster than the generative baseline, which decomposes into $\sim 18\times$ from eliminating autoregressive decoding (359.1 to 19.5 ms at depth 12) and $\sim 3.8\times$ from pruning (19.5 to 5.1 ms); the conversion, not the pruning, dominates. It matches encoder intent accuracy and cannot emit a malformed output. The generative baseline is unconstrained; constrained decoding would reduce its parse failures at additional decode cost.

Model	Intent Acc	Slot F1	Parse-fail	P50 (ms)
Generative SLM (gpt2, autoregressive)	48.3	44.6	46.7	359
Encoder discriminative (DistilBERT)	98.98	95.1	0	7.0
Ours: probe-pruned (gpt2 depth-3)	97.95	91.0	0	5.1

5 RESULTS

5.1 Probe-guided candidate pruning (C1). Over the predefined sparse grid $\{3, 6, 9, 12\}$ the ATIS frozen probe scores 95.0/95.4/95.3/96.0%; the $\epsilon=0.02$ rule selects depth 3 (within 1pt of full). All five datasets independently return depth 3 on this grid. Fine-tuning at depth 3 matches or beats full depth on intent accuracy on ATIS/SNIPS while being $\sim 4\times$ faster. To test the probe as a *predictor* rather than just a decision rule on this grid, we compute Spearman’s rank correlation between frozen probe accuracy and fine-tuned intent accuracy *within each dataset* (the metric aligned with C1). Across the six evaluated depths $\{1, 2, 3, 6, 9, 12\}$ per dataset, per-dataset ρ is -0.612 (ATIS), $+0.353$ (SNIPS), -0.754 (MASSIVE), -0.486 (CLINC150), -0.771 (BANKING77); mean -0.454 , median -0.612 . **The frozen intent probe does not rank-order depths within a dataset.** The pooled cross-dataset $\rho = +0.871$ we report elsewhere reflects between-dataset difficulty (easier tasks have higher probe *and* higher fine-tuned accuracy), not within-dataset depth ranking. Reported honestly, frozen intent probes reliably identify an *aggressive candidate region* for pruning but do not reliably rank fine-tuned depths within individual datasets—the final deployed depth is confirmed by downstream fine-tuning.

Does linear accessibility imply fine-tuning sufficiency? No. The dense frozen probe of layers 1–12 (Table 4) shows intent representations are already highly linearly accessible at *layer 1* on every dataset (ATIS 95.2%, SNIPS 96.7%, MASSIVE 74.4%, CLINC150 87.6%, BANKING77 81.9%). Under the $\epsilon=0.02$ rule the dense sweep would select depth 1 for all five datasets. Yet downstream fine-tuning at depth 1 (Table 5) loses 2.7–4.1pt of intent accuracy on the fine-grained benchmarks and substantially more slot F1 (up to 10.9pt on SNIPS). On ATIS and SNIPS, depth-1 intent accuracy is at most 3.3pt below depth 3. Depth 2 recovers much of this gap—within 1.73pt of depth 3 on intent across all five datasets—while depth 3 is the shallowest *evaluated* fine-tuned depth (among $\{1, 2, 3, 6, 9, 12\}$) that consistently preserves both intent accuracy and slot F1. Frozen intent probes over-report depth-1’s usability; the substantive negative finding is that linear accessibility of

Table 3: Dense per-layer token-level BIO-tag probe (frozen backbone; linear probe validation accuracy %). ATIS and MASSIVE saturate at L1–L2 and are flat thereafter; SNIPS shows a +7pt L1→L2 jump that mirrors the downstream depth-1 slot F1 degradation in Table 5. ATIS and MASSIVE do not exhibit an analogous L1→L2 transition.

Dataset	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12
ATIS	95.4	95.6	96.1	96.4	96.4	96.4	96.3	96.2	95.6	95.7	95.6	95.5
SNIPS	77.4	84.5	84.9	84.7	85.6	84.9	81.4	84.8	85.0	84.6	83.5	84.6
MASSIVE	82.9	83.2	82.2	83.2	82.9	82.5	82.9	82.1	82.6	82.5	81.5	83.8

Table 4: Dense per-layer frozen probe (intent accuracy %), all five datasets, layers 1–12. Bold denotes the maximum probe accuracy within each dataset (ties bolded); layer 1 is competitive or best on every dataset.

Dataset	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12
ATIS	95.2	95.2	95.1	95.6	95.6	94.9	95.9	95.1	95.1	96.2	95.2	95.9
SNIPS	96.7	96.9	96.1	96.5	97.0	97.0	96.7	97.0	96.7	96.7	96.6	94.9
MASSIVE	74.4	75.0	75.0	74.1	75.5	73.5	74.8	74.3	72.3	72.3	72.1	72.9
CLINC150	87.6	85.6	86.8	87.6	86.9	87.1	85.8	86.9	84.6	85.0	85.5	86.1
BANKING77	81.9	79.9	80.5	79.8	76.9	78.5	79.3	79.1	77.5	76.2	73.5	75.7

task information precedes fine-tuning sufficiency at the extreme shallow end, and cheap intent-only probes cannot detect the gap.

Slot probes reveal what intent probes miss (on SNIPS). We repeat the dense layer sweep with a token-level BIO-tag probe (frozen backbone, linear tag classifier per layer) on the three slot-bearing datasets (Table 3). ATIS and MASSIVE slot signal is nearly flat with depth (within ~1pt from L1 to L12), yet their downstream depth-1 slot F1 still drops by 4.5 and 2.9pt respectively—so the flat slot probe on these two does *not* explain their smaller collapses. SNIPS, in contrast, shows a sharp +7pt jump from L1 (77.4%) to L2 (84.5%), then plateaus; that L1→L2 transition mirrors the downstream depth-1 SNIPS slot F1 collapse (82.2 at depth 3 → 71.3 at depth 1). We take this as single-dataset evidence that *task-aligned* token probes can reveal a phase transition that intent-only probing hides, and as motivation for a slot-probe-aware selection criterion: pick the shallowest layer that jointly satisfies the intent- and slot-probe ϵ -rules. On SNIPS that refinement would select depth 2 instead of depth 1 and restore method-experiment consistency; we present it as a promising refinement of C1 rather than the paper’s headline rule.

Matched budget control. To isolate pruning depth from training budget, we train depths 1, 3, and 12 on the same 3,000 ATIS examples. Depths 3 and 12 achieve identical intent accuracy of 96.08%, while P50 latency falls from 20.4 to 5.4 ms. Depth 1 reaches 92.83% intent accuracy at 1.8 ms, confirming that the extreme shallow setting introduces a measurable capacity cost even at matched budget.

5.2 Reliability (C2). Generative parse-failure rate tracks slot-schema complexity: 46.7% on ATIS (135 BIO tag labels; the M1 baseline run itself used 101 under a pre-fix tag-vocabulary convention, see Table 1 footnote), 3.3% on SNIPS (72), 0% on intent-only CLINC150; generative latency likewise scales with output length (359/243/83 ms). The discriminative model is flat (~5–7 ms) and always valid, independent of schema complexity.

5.3 CRF ablation. With a length-normalized CRF loss, the CRF slot head gives a small consistent gain over softmax at depth 3 (ATIS +0.19, SNIPS +0.27, MASSIVE +0.09 F1); it does not close the gap to the bidirectional encoder, suggesting that transition modeling alone does not explain the gap. Attention directionality and other encoder-specific properties (pretraining objective, tokenizer, model family) remain possible explanations.

Slot-loss weight λ . We sweep $\lambda \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$ on ATIS depth-3 (Table 8). Intent accuracy is flat within 0.5pt (97.61–98.12); slot F1 monotonically rises with λ from 89.56 to 91.53 as expected. The paper’s default $\lambda=2$ sits in the middle of a smooth curve—no cliff, no sharp optimum. The main conclusions are insensitive to λ within this range.

Table 5: Depth-1 vs. depth-2 vs. depth-3 fine-tuning under the matched training-data protocol, capped at 5,000 examples per dataset (seed 42). Depth 2 is within 1.73pt of depth 3 on intent across all five datasets; SNIPS slot F1 is the only sizable residual gap. Frozen-probe accessibility at layer 1 does not imply fine-tuning sufficiency at depth 1.

Dataset	d1 Int	d2 Int	d3 Int	Δ_{3-1}	d1 SlotF1	d2 SlotF1	d3 SlotF1	d1 P50	d2 P50	d3 P50
ATIS	94.71	97.78	97.95	-3.24	86.46	90.74	90.96	1.94	3.26	5.12
SNIPS	97.71	98.14	98.00	-0.29	71.27	78.26	82.17	1.72	3.19	4.87
MASSIVE	69.70	72.46	73.81	-4.11	47.66	51.32	50.59	1.15	1.84	2.65
CLINC150	75.87	77.29	79.02	-3.15		intent-only		1.19	1.81	2.64
BANKING77	75.68	77.69	78.34	-2.66		intent-only		1.20	1.81	2.91

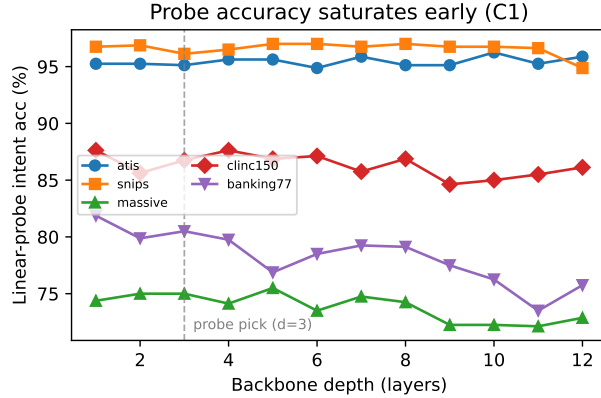


Figure 2: Frozen intent probe accuracy across backbone depth. On the predefined sparse candidate grid, probe performance is already saturated at depth 3, motivating depth 3 as an aggressive candidate pruning point before fine-tuning. The dense diagnostic sweep shows that linear accessibility emerges even earlier, but downstream experiments (Section 5) demonstrate that accessibility alone does not imply fine-tuning sufficiency.

5.4 Fair head-to-head. Under the matched training-data protocol, capped at 5,000 examples per dataset, ours (depth 3) vs. DistilBERT: ATIS 97.95 vs 97.95 intent (tie), 91.0 vs 94.5 slot F1; the encoder leads on slots and on fine-grained intent (CLINC150 82.0 vs 92.96). Our recipe’s value is converting an existing decoder into a fast discriminative model, not beating a purpose-built encoder.

6 NEGATIVE RESULTS: CPU INFERENCE OPTIMIZATIONS

Table 9 summarizes four conditions on the same weights. **Dynamic INT8** gives no benefit: at depth 3 the Linear layers are not the bottleneck, per-batch quant/dequant overhead offsets the cheaper matmuls, and the path silently fails to run on a stock CPU build without `torch.backends.quantized.engine = "qnnpack"`—a portability footgun. **torch.compile** is marginally *slower*: compile overhead and dynamic sequence shapes prevent graph fusion from amortizing at this scale. **ONNX Runtime** yields a real 4.5× speedup at identical accuracy, but the modern dynamo exporter fails on the decoder and the legacy exporter rejects the dataclass output; it only succeeds after wrapping the model to return a plain `(intent_logits, slot_logits)` tuple. Takeaway: for shallow closed-set NLU the win came from *pruning + discriminative single-pass*, not post-hoc kernel/quant tricks; ONNX graph execution is an orthogonal, stackable win with real export friction.

Implicit-slot scope (measured, not estimated). A slot value is *implicit* if it is not a contiguous span of the utterance; span-based taggers cannot recover it by construction. We measured the implicit-slot rate on all three slot-filling benchmarks and found it is exactly **0%** (ATIS: 0/16,350; SNIPS: 0/37,542; MASSIVE: 0/14,159 gold slot values). These benchmarks are span-annotation-first by design, so our span-based heads have no theoretical recall ceiling here; tasks with a genuine implicit-

Table 6: Multi-seed intent accuracy (mean $\pm$ std, 3 seeds; MASSIVE 2 seeds). With only 3 seeds we report descriptive statistics rather than an inferential test: no clear separation between depth-3 and depth-12 is observed on ATIS/SNIPS/CLINC150, while MASSIVE and BANKING77 show a small, seed-robust advantage for depth 12. Every depth-3 config is $\sim 4\times$ faster.

Dataset	depth 3	depth 12	separation across 3 seeds?
ATIS	97.95 $\pm$ 0.14	97.84 $\pm$ 0.29	none observed
SNIPS	98.19 $\pm$ 0.18	98.14 $\pm$ 0.00	none observed (d3 nominally higher)
CLINC150	78.06 $\pm$ 0.91	80.44 $\pm$ 1.68	none observed
MASSIVE	73.49 $\pm$ 0.32	74.50 $\pm$ 0.12	~ 1 pt, seed-robust
BANKING77	78.74 $\pm$ 0.29	81.32 $\pm$ 0.81	~ 2.6 pt, seed-robust

Table 7: Depth Pareto (ours). Accuracy peaks around depth 6–9 then plateaus; latency is $\sim$ linear in depth. Intent accuracy / slot F1 / P50 ms.

Dataset	depth 3	depth 6	depth 9	depth 12
ATIS	97.95/91.0/5.1	97.78/92.3/10.9	98.12/92.2/15.8	97.44/91.9/19.5
SNIPS	98.00/82.2/4.9	98.29/84.5/9.5	98.43/84.9/15.7	98.14/84.7/18.9
MASSIVE	73.81/50.6/2.7	74.98/55.7/5.3	75.62/55.1/7.6	74.61/54.3/10.5
CLINC150	79.02/-/2.6	81.44/-/5.1	81.93/-/7.7	82.00/-/10.1
BANKING77	78.34/-/2.9	80.91/-/5.4	81.79/-/9.1	80.78/-/10.7

slot rate (e.g. coreference- or ellipsis-dependent slot filling) would need a generative fallback or explicit non-span recovery rules, which we leave to future work rather than force onto benchmarks that do not exhibit the phenomenon. This is a scope-validation result, not an independent contribution.

7 DISCUSSION AND LIMITATIONS

Closed-set assumption; open-set/novel-intent detection is out of scope. Generation is still needed for free-form tasks. The representation accessibility-versus-fine-tuning-sufficiency result is established here for a single 124M causal decoder on a single CPU. **This is the principal limitation of the paper.** We deliberately state the scope narrowly: we do not claim that depth 3 is the right operating point for decoders in general, nor that the accessibility-versus-sufficiency gap appears at the same depth—or at all—in modern 2–7B decoders, whose depth, width, and pretraining differ substantially from GPT-2. Establishing whether the phenomenon is a property of causal decoders broadly or an artifact of this backbone requires multi-backbone replication on accelerator hardware, which our CPU-only setting could not support and which we leave to future work. Readers should treat C1 as a procedure validated on one backbone and the accessibility-versus-sufficiency finding as a single-model existence result: it demonstrates the failure mode is real, not that it is universal. The fine-grained gap on BANKING77 shows depth genuinely helps for many same-domain intents—probe-guided candidate pruning trades a small, quantified accuracy cost there for a $4\times$ speedup.

Fairness caveats on baselines. Our generative baseline is bounded to 3,000 train / 300 eval examples because autoregressive CPU decoding is expensive; the discriminative rows use the matched training-data protocol described in Section 4, capped at 5,000 examples per dataset. The latency and parse-failure comparisons remain apples-to-apples (they are architecture properties). The matched budget control in Section 5 isolates pruning depth from training budget: depths 3 and 12 reach identical intent accuracy at 3,000 examples, confirming that pruning depth-12 $\rightarrow$ 3 is an architecture win, not a training-budget confound.

8 CONCLUSION

Generate less, classify more: cheap frozen probes identify an aggressive candidate pruning region, downstream fine-tuning confirms a shallow operating depth, and discriminative heads deliver fast, reliable, single-pass intent and slot prediction from an existing generative SLM.

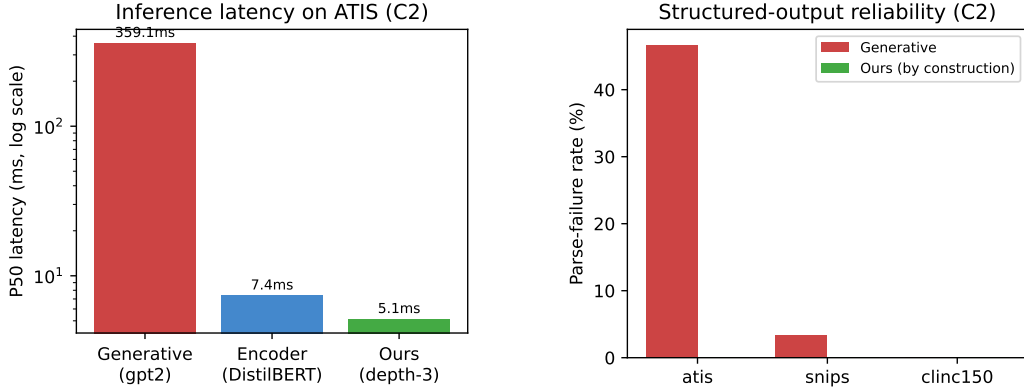


Figure 3: Left: P50 CPU latency on ATIS (log scale). Right: structured-output parse-failure rate; a discriminative head over a closed label set has no malformed-output mode, so this is a property of the output parameterization rather than an empirical finding.

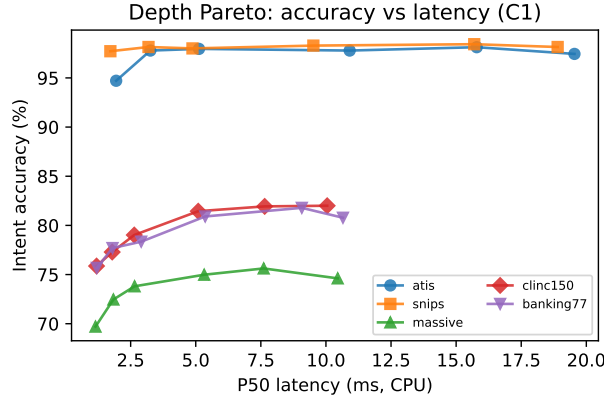


Figure 4: Depth Pareto: intent accuracy vs. P50 latency for depths $\{3, 6, 9, 12\}$, all five datasets. Accuracy gains beyond depth 6–9 are small relative to the added latency.

REPRODUCIBILITY STATEMENT

All datasets are public. Code, exact commands, model revisions, seeds, and the CPU specification are released via an anonymized repository. Every table is regenerated by a single aggregation script from per-run JSON.

AI-USE STATEMENT

The authors used an AI coding assistant to help implement experiment scaffolding, run the training/evaluation harness, and draft prose. All experimental design, result interpretation, and claims were verified by the authors; all reported numbers were produced by the released code on public benchmarks.

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Table 8: Slot-loss weight λ sweep on ATIS depth-3 (softmax slot head, seed 42). Intent accuracy is stable; slot F1 rises monotonically with λ ; the paper’s default $\lambda=2$ is unremarkable within the sweep.

λ	Intent Acc	Slot F1	P50 (ms)
1.0	98.12	89.56	4.89
1.5	97.95	90.68	5.07
2.0 (default)	97.95	90.96	4.98
2.5	97.78	91.23	5.18
3.0	97.61	91.53	6.72

Table 9: CPU inference optimizations on the depth-3 ATIS model (identical weights across conditions; P50 single-example latency, Apple Silicon arm64, 10 threads). Full detail and root-cause analysis in RESULTS_M5_negresults.md.

Condition	Intent Acc	Slot F1	P50 (ms)	vs FP32	Verdict
FP32 baseline	97.95	90.96	5.13	1.00×	reference
Dynamic INT8 (qnnpack)	97.95	90.68	5.02	1.02×	no benefit
<code>torch.compile</code> (Ansel et al., 2024)	97.95	90.96	5.39	0.95×	<i>slower</i>
ONNX Runtime (ONNX Runtime developers, 2019)	97.95	90.96	1.14	4.5×	big win, brittle export

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